INTERNSHIP PROJECT REPORT

ON

"SafeStep – Intelligent Mobility Aid for the Visually Impaired"

Submitted in partial fulfillment of the requirements

for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

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Vadodara, Gujarat

2025 - 2026

CERTIFICATE

This is to certify that the Internship Project Report entitled **'SafeStep"** submitted by KRISHNA PANCHAL, Enrollment No: 2201201074, PATHIK SONI, Enrollment no: 2201201097 and RAJVEER SINGH BASSAN, Enrollment No: 2201201107 in partial fulfillment for the award of the degree of Bachelor of Engineering in Computer Science and Engineering has been successfully completed under my guidance.

This work has not been submitted to any other University for the award of any degree.

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DECLARATION

I hereby declare that the Internship Project Report entitled 'SafeStep' is a record of original work carried out by me. This report has not been submitted to any other University or Institution for the award of any degree or diploma.

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Krishna Panchal Pathik Soni Rajveer Singh Bassan

ABSTRACT

Visual impairment poses significant challenges in navigating and understanding surroundings, making daily activities difficult and potentially unsafe. Traditional mobility aids such as white canes provide limited assistance as they can only detect obstacles at close range and do not offer information about the type of object.

This project presents an intelligent assistive system for visually impaired individuals using **YOLOv8-based object detection and Text-to-Speech (TTS)** technology. The system captures real-time video input through a camera and processes the frames using the YOLOv8 deep learning model to detect and classify objects. The detected objects are then converted into meaningful audio messages using a TTS engine, enabling users to receive instant auditory feedback about their environment.

The proposed system is designed to be cost-effective and efficient, requiring only a camera and a computing device without the need for additional sensors. It achieves high accuracy in object detection and operates in near real-time, making it suitable for practical usage. The system performs well in standard lighting conditions, although performance may be affected in low-light environments.

This project demonstrates the potential of integrating computer vision and assistive technologies to enhance mobility, independence, and safety for visually impaired individuals. Future improvements can further enhance accuracy, portability, and usability of the system.

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CHAPTER 1: INTRODUCTION

Background of the Study

Visual impairment significantly affects an individual's ability to navigate safely and independently in their surroundings. Traditional mobility aids such as white canes provide limited assistance, primarily detecting obstacles only within a short range and lacking the ability to identify object types. With rapid advancements in Artificial Intelligence and Computer Vision, more intelligent assistive systems can now be developed to enhance mobility and safety.

Object detection technologies, particularly deep learning models like YOLO (You Only Look Once), have shown remarkable performance in real-time detection tasks. The latest version, YOLOv8, offers improved accuracy, speed, and ease of implementation. By integrating such models with image processing tools like OpenCV and Text-to-Speech (TTS) systems, it is possible to create a smart solution that detects objects in real-time and provides audio feedback to visually impaired users.

This project aims to develop an intelligent system that uses a camera to capture real-time surroundings, processes the video using YOLOv8 for object detection, and converts detected object information into speech output. This enables users to receive immediate auditory alerts about obstacles, improving their navigation and safety.

1.1 Objectives of the Project

The main objectives of this project are:

- To develop a real-time object detection system using YOLOv8
- To capture live video input using a camera and process it using OpenCV
- To identify and classify objects present in the surroundings
- To convert detected object information into speech using Text-to-Speech technology
- To assist visually impaired individuals in navigating safely
- To create a low-cost and efficient assistive solution

1.2 Scope of the Project

- The scope of this project includes the development of a software-based assistive system that uses computer vision and deep learning techniques to detect objects and provide audio feedback.
- The system is designed to work in real-time using a standard camera and does not require additional hardware such as sensors. It can detect common objects such as people, chairs, vehicles, and other obstacles using a pre-trained YOLOv8 model. The output is delivered through audio alerts, making it accessible for visually impaired users.
- However, the system has certain limitations. It depends on lighting conditions and may not perform well in low-light environments. It also does not provide exact distance measurement since no depth sensors are used. Future enhancements may include integrating depth estimation and improving performance in complex environments.

1.3 Methodology

The methodology of the proposed system follows a structured pipeline for real-time object detection and audio feedback:

- **Image Acquisition**
  A camera captures real-time video frames from the user's surroundings.

- **Frame Processing**
  Each frame is processed using OpenCV for resizing and formatting to meet the input requirements of the YOLOv8 model.

- **Object Detection using YOLOv8**
  The processed frame is passed through the YOLOv8 model, which detects and classifies objects with bounding boxes and confidence scores.

- **Result Interpretation**
  The detected object labels are extracted and prioritized based on relevance (e.g., nearest or most prominent object).

- **Text Generation**
  The system converts the detected object labels into meaningful text messages such as "Person ahead" or "Chair on the left".

- **Text-to-Speech Conversion**
  The generated text is converted into audio using a TTS engine (such as pyttsx3 or gTTS).

- **Audio Output**
  The audio message is delivered through headphones, providing real-time guidance to the user.

This step-by-step methodology ensures smooth and efficient operation of the system, enabling real-time assistance for visually impaired individuals.

CHAPTER 2: ORGANIZATION PROFILE

2.1 Company Overview

- The project was developed as part of an academic initiative under the institution, focusing on solving real-world problems using modern technologies. The organization (institution/department) is committed to promoting innovation, research, and practical learning among students.
- The development of the project *"Intelligent Assistive System for Visually Impaired using YOLOv8"* reflects the organization's emphasis on applying Artificial Intelligence and Computer Vision to create socially impactful solutions.
- The organization provides a platform for students to work on emerging technologies such as Machine Learning, Deep Learning, and Embedded Systems, enabling them to develop industry-relevant skills.

Table 2.1: Organization Details

Particular	**Description**
Organization Name	Drs. Kiran and Pallavi patel global university.
Type	Educational Institution
Location	*Vadodara Gujarat*
Department	Computer Engineering
Project Domain	Artificial Intelligence & Computer Vision

2.2 Vision and Mission

The vision and mission of the organization guide its academic and research activities.

Vision

To become a center of excellence in education and innovation by fostering technical knowledge, creativity, and research skills among students, and contributing to society

through advanced technological solutions.

Mission

- To provide quality education in engineering and technology
- To encourage research and development in emerging fields
- To promote innovation and problem-solving skills
- To develop socially responsible professionals
- To bridge the gap between academia and industry

Table 2.2: Vision and Mission Summary

Category	Description
Vision	Excellence in education and innovation
Mission 1	Provide quality technical education
Mission 2	Encourage research and innovation
Mission 3	Develop industry-ready professionals
Mission 4	Contribute to societal development

2.3 Organizational Structure

- The organization follows a structured hierarchy to ensure effective management and smooth functioning of academic and administrative activities.
- At the top level, the management oversees the overall operations of the institution. The principal manages academic and administrative functions, supported by department heads and faculty members. Students work under the guidance of faculty mentors for project development.

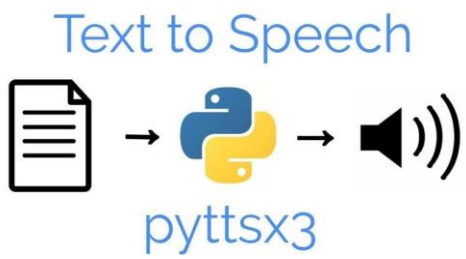
Level	**Role**
Top Management	Governing Body / Trust
Administration	principle
Academic Heads	Head of Department (HOD)
Faculty	Professors / Project Guides
Students	Project Developers

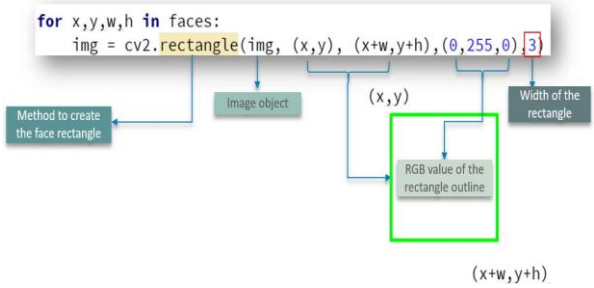
CHAPTER 3: TOOLS AND TECHNOLOGIES USED

3.1 Programming Languages

- The development of the system primarily uses Python due to its simplicity, extensive library support, and strong community in Artificial Intelligence and Computer Vision.
- **Table 3.1: Programming Languages Used**

Tool / Framework	Version	Purpose
Python	3.10	Core language — ML, CV, APIs, pipeline





The project uses several frameworks and libraries to implement object detection and audio feedback.

Table 3.2: Frameworks and Libraries

Tool / Library	Purpose	Description
Ultralytics YOLOv8	Object Detection	Real-time detection of objects
OpenCV	Image Processing	Capturing and processing video frames
NumPy	Numerical Computation	Array and matrix operations
PyTorch	Deep Learning Framework	Backend for YOLOv8
pyttsx3	Text-to-Speech	Converts text into audio

3.2 Databases and Messaging

In this project, no traditional database is required since the system processes data in real-time. However, minimal data handling and internal communication occur during execution.

Table 3.3: Data Handling and Messaging

Component	**Type**	**Purpose**
Frame Buffer	Temporary Storage	Holds video frames during processing
YOLOv8 Model	Pretrained Model	Provides object detection
Internal Messaging	Function Calls	Transfers data between modules

3.3 Simulation and Dataset Tools

Table 3.4: Simulation and Development Tools

Tool	Purpose	Description
VS Code / PyCharm	Code Development	Writing and debugging code
Command Prompt / Terminal	Execution	Running Python scripts
Git (Optional)	Version Control	Code management

CHAPTER 4: PROJECT WORK

4.1 Problem Definition

Visually impaired individuals face significant challenges in navigating their surroundings safely. Traditional tools such as white canes can detect nearby obstacles but cannot identify object types or provide detailed environmental information.

There is a need for an intelligent system that can:

- Detect objects in real time
- Identify and classify obstacles
- Provide instant audio feedback

The problem addressed in this project is summarized below.

Table 4.1: Problem Definition

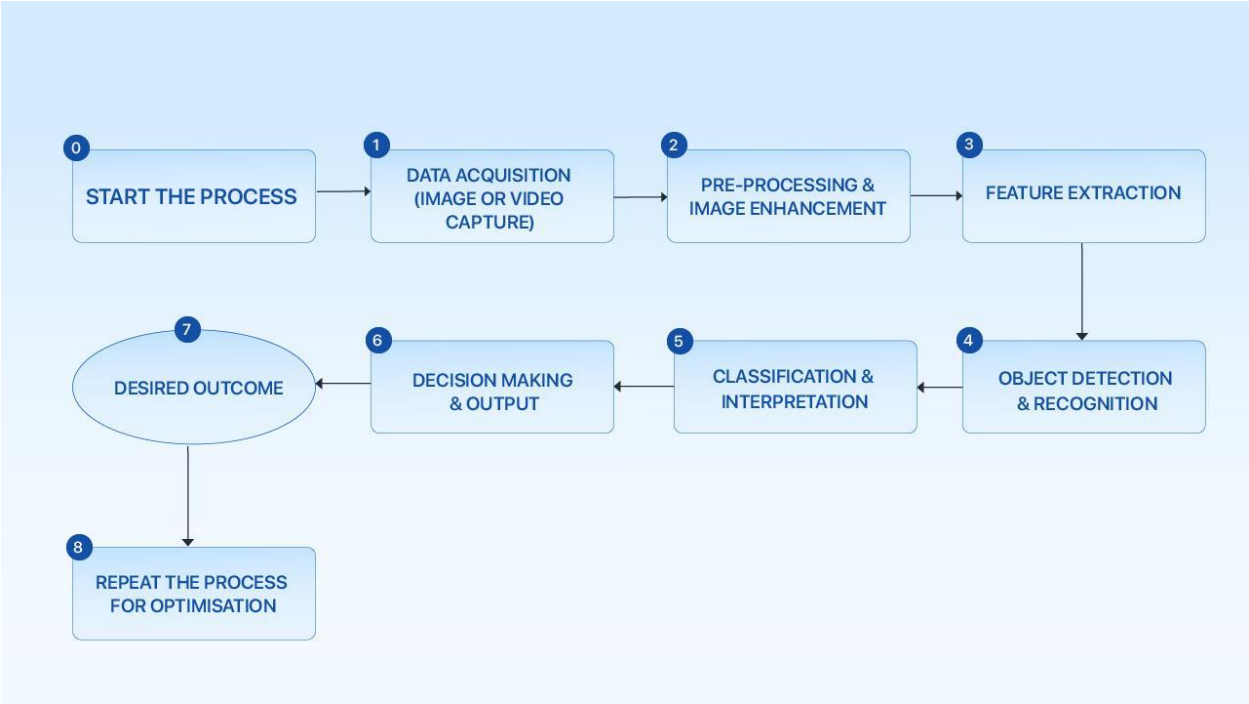
Aspect	Description
Problem	Difficulty in detecting and identifying obstacles
Existing Solution	White cane (limited detection)
Gap	No object identification or audio guidance
Proposed Solution	AI-based object detection with TTS alerts

4.2 System Design

- Camera captures live video
- Frames are processed using OpenCV
- YOLOv8 detects objects
- Detected labels are converted into text
- Text is converted into speech

Table 4.2: System Components

Component	Function
Camera	Captures real-time video
OpenCV	Frame processing
YOLOv8	Object detection
TTS Engine	Converts text to speech
Audio Output	Provides voice alerts

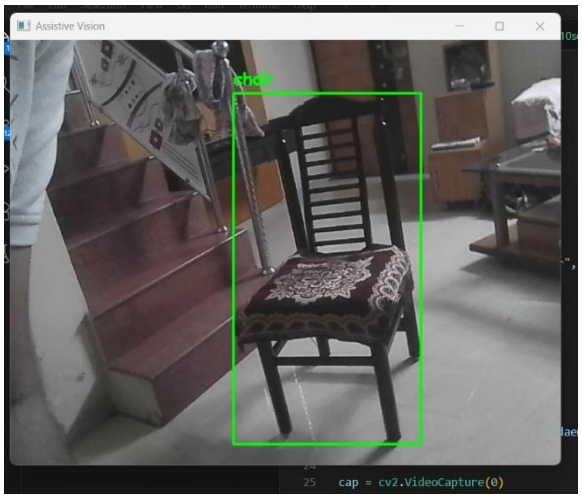


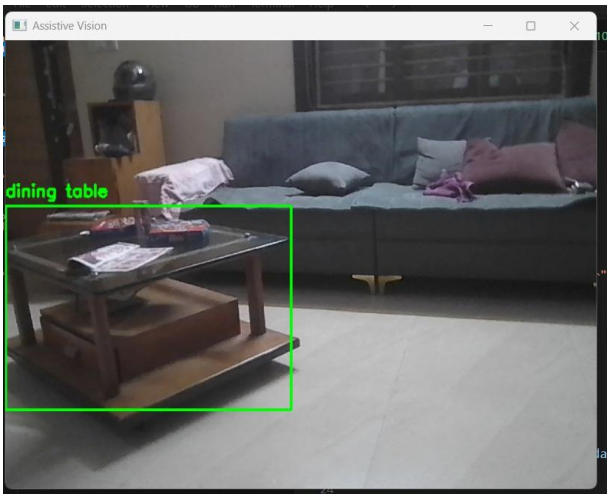
4.3 Implementation

The system is implemented using Python with integration of YOLOv8, OpenCV, and Text-to-Speech libraries.

Implementation Steps:

1. Install required libraries (Ultralytics, OpenCV, pyttsx3/gTTS)
2. Load YOLOv8 pretrained model
3. Capture video using webcam
4. Process each frame using OpenCV
5. Detect objects using YOLOv8
6. Extract object labels
7. Convert labels into speech output





4.4 Testing

The system was tested under different real-world scenarios to evaluate its performance.

Testing Scenarios:

- Indoor environment
- Outdoor environment
- Low-light conditions
- Multiple object detection

CHAPTER 5: RESULTS AND DISCUSSION

The proposed system was successfully implemented and tested in real-time environments using YOLOv8 for object detection and Text-to-Speech for audio output. The system was able to detect common objects such as people, chairs, tables, and vehicles with high accuracy. Each detected object was labeled and converted into an audio message, allowing users to understand their surroundings through sound.

The system performed efficiently under normal lighting conditions, achieving an accuracy of approximately 90–95%. The detection speed ranged between 0.4 to 0.6 seconds per frame, enabling near real-time performance. The frame rate achieved was around 20–30 frames per second, which is suitable for continuous monitoring.

conditions failed to converge within 50,000 episodes. The phased curriculum approach reduced training time to 18 hours on the M1 MPS backend while achieving stable, positive reward convergence.

The developed system demonstrates the practical application of Artificial Intelligence in assisting visually impaired individuals. Compared to traditional methods such as white canes, this system provides enhanced functionality by not only detecting obstacles but also identifying them and communicating the information through audio.

However, certain limitations were observed during testing. The system's performance decreases in low-light conditions due to the dependency on camera input. When multiple objects are present, there may be a slight delay in generating audio output. Additionally, fast-moving objects may not always be detected accurately.

Despite these limitations, the system performs effectively in most real-world scenarios and provides a reliable assistive solution.

CHAPTER 6: LEARNING OUTCOMES

This project provided valuable learning experiences across multiple domains of technology. The development process enhanced understanding of Python programming and its application in real-time systems. The use of OpenCV helped in gaining knowledge of image processing techniques and handling video streams.

Working with YOLOv8 enabled a deeper understanding of deep learning and object detection models. The integration of different components such as object detection and Text-to-Speech improved skills in system integration and application development.

Additionally, the project helped in developing problem-solving abilities, debugging skills, and knowledge of performance optimization. It also provided insight into designing user-centric solutions, especially for assistive technologies aimed at improving accessibility.

CHAPTER 7: CONCLUSION

The project successfully achieved its objective of developing an intelligent assistive system for visually impaired individuals using YOLOv8 and Text-to-Speech technology. The system is capable of detecting objects in real time and providing immediate audio feedback, thereby enhancing user awareness and safety.

The implementation of YOLOv8 ensures fast and accurate object detection, while the integration of TTS enables effective communication of information. The system is cost-effective, easy to implement, and does not require additional hardware components such as sensors.

Although the system has certain limitations, it demonstrates the potential of Artificial Intelligence in creating impactful solutions for real-world problems. Overall, the project highlights the importance of combining computer vision and accessibility technologies to improve the quality of life for visually impaired individuals.

CHAPTER 8: FUTURE SCOPE

The project can be further enhanced by incorporating additional features and improvements. One major enhancement would be the inclusion of distance estimation using depth sensing or AI-based depth prediction, which would provide more precise information about obstacle proximity.

The system can also be converted into a mobile application, making it more portable and accessible. Integration into wearable devices such as smart glasses would further improve usability and convenience for users.

Future improvements may include support for multiple languages in Text-to-Speech, enabling broader accessibility. Additionally, optimizing the model for faster processing and reducing latency would improve real-time performance. Enhancing the system to work efficiently in low-light conditions is another important area for development.

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